Glycolysis: Complete Step-by-Step Biochemical Reference

Enzymes, Reaction Equations, Carbon Skeletons, Phosphate Transfers, and Electron Movement

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1 Overview of the Glycolytic Pathway

Glycolysis is a ten-reaction metabolic pathway that converts one six-carbon glucose molecule into two three-carbon pyruvate molecules.

All ten reactions occur in the cytosol.

The pathway is divided into two phases:

Phase 1. Energy-investment phase: Steps 1–5.

Two ATP molecules are consumed. The six-carbon sugar is phosphorylated, rearranged, and cleaved into two three-carbon molecules.

Phase 2. Energy-payoff phase: Steps 6–10.

Two glyceraldehyde-3-phosphate molecules are oxidized and converted into pyruvate. Four ATP and two NADH molecules are produced.

1.1 Whole-pathway diagram

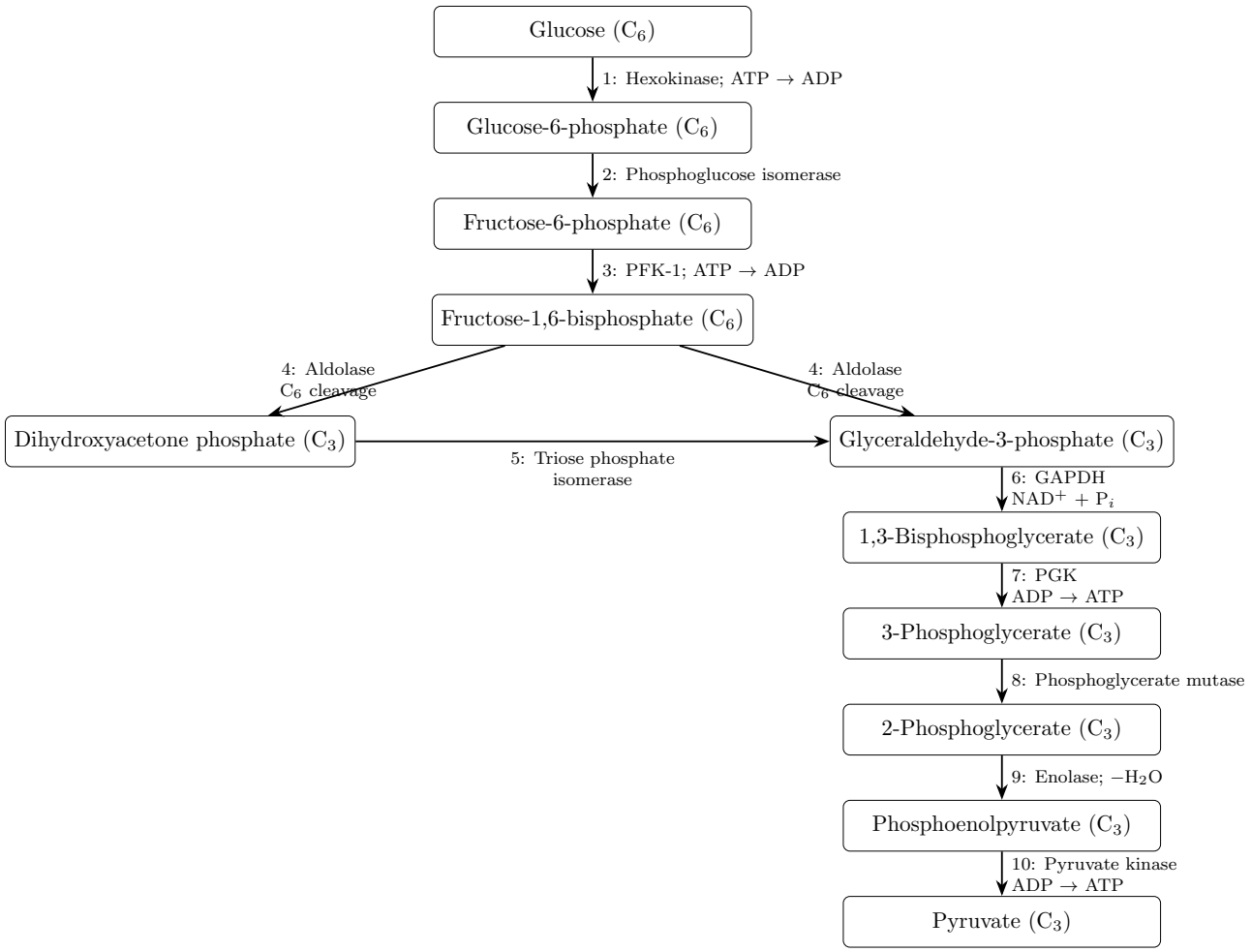


Figure 1: The ten reactions of glycolysis. After step 5, two glyceraldehyde-3-phosphate molecules proceed through each of steps 6–10.

Important stoichiometric convention: Steps 1–5 occur once per original glucose molecule. Steps 6–10 occur twice per glucose molecule because two glyceraldehyde-3-phosphate molecules are formed.

2 Structural Notation and Atom Tracking

2.1 Abbreviations

Notation	Meaning
G6P	Glucose-6-phosphate
F6P	Fructose-6-phosphate
FBP	Fructose-1,6-bisphosphate
DHAP	Dihydroxyacetone phosphate
GAP or G3P	Glyceraldehyde-3-phosphate
1,3-BPG	1,3-Bisphosphoglycerate
3-PG	3-Phosphoglycerate
2-PG	2-Phosphoglycerate
PEP	Phosphoenolpyruvate
P_i	Inorganic phosphate
ATP	Adenosine triphosphate
ADP	Adenosine diphosphate
NAD^+	Oxidized electron acceptor
NADH	Reduced electron carrier
PFK-1	Phosphofructokinase-1
GAPDH	Glyceraldehyde-3-phosphate dehydrogenase
PGK	Phosphoglycerate kinase

2.2 Carbon numbering

Carbon atoms are numbered according to their positions in the intermediates, not according to a universal numbering system maintained unchanged through every reaction.

- Glucose contains six carbon atoms, numbered C1–C6.
- The isomerization in step 2 converts an aldose into a ketose.
- Aldolase cleaves the six-carbon intermediate between C3 and C4.
- The resulting triose phosphates each contain three carbon atoms.
- From step 6 onward, each triose molecule follows the same five-reaction sequence.

The original glucose carbon atoms are conserved: glycolysis does not release carbon dioxide.

2.3 Phosphate notation

In structural formulas, the phosphate ester group is abbreviated as

$$-\mathrm{O}-\mathrm{PO}_3^{2-}.$$

This notation shows the essential connectivity between the sugar oxygen and phosphorus. It suppresses some protonation details because the actual charge distribution depends on pH and the ionic environment.

The phosphate group donated by ATP is transferred to the sugar through an oxygen atom. It is not transferred as an isolated phosphorus atom.

3 Step 1: Phosphorylation of Glucose

3.1 Reaction

$$\boxed{\text{Glucose} + \text{ATP} \xrightarrow[\text{Mg}^{2+}]{\text{Hexokinase}} \text{Glucose}-6\text{-phosphate} + \text{ADP}}$$

Enzyme: Hexokinase. In liver and pancreatic beta cells, glucokinase (hexokinase IV) performs the corresponding reaction.

3.2 Reactants and products

- Reactants: glucose, ATP.
- Products: glucose-6-phosphate, ADP.
- Cofactor: usually Mg^{2+}, which helps coordinate nucleotide phosphate groups.
- ATP consumed: one molecule per glucose.

3.3 Structural representation

$$HOCH_2-(CHOH)_4-CHO \quad \text{(open-chain glucose, schematic)}$$

$$\text{Glucose} + \text{ATP} \longrightarrow \text{Glucose}-6\text{-phosphate} + \text{ADP}$$

In the open-chain representation, the terminal C6 hydroxyl group becomes a phosphate ester:

$$-CH_2OH \longrightarrow -CH_2OPO_3{}^{2-}.$$

3.4 Bond and atom changes

1. The C6 hydroxyl oxygen of glucose acts as the acceptor atom for the phosphoryl group.
2. The terminal phosphate of ATP is transferred to this oxygen.
3. The C6 oxygen forms a bond to phosphorus.
4. ATP becomes ADP after losing its terminal phosphoryl group.
5. The six-carbon skeleton of glucose remains intact.

3.5 Electron movement

The hydroxyl oxygen is the nucleophile in the phosphoryl-transfer reaction. It attacks the terminal phosphorus of ATP, while the phosphoryl-transfer transition state is stabilized by the enzyme and Mg^{2+}.

This is a nucleophilic substitution at phosphorus, not oxidation of the glucose carbon skeleton.

3.6 Energy and physiological significance

One ATP is consumed. The resulting glucose-6-phosphate is charged and does not readily cross the plasma membrane through glucose transporters. Phosphorylation also prepares glucose for subsequent metabolic reactions.

Reference: Step 1. The phosphate is attached to C6, not C1. No NADH is produced.

4 Step 2: Isomerization of Glucose-6-phosphate

4.1 Reaction

$$\boxed{\text{Glucose}-6\text{-phosphate} \xrightleftharpoons{\text{Phosphoglucose isomerase}} \text{Fructose}-6\text{-phosphate}}$$

Enzyme: Phosphoglucose isomerase, also called glucose-6-phosphate isomerase.

4.2 Reactants and products

- Reactant: glucose-6-phosphate.
- Product: fructose-6-phosphate.
- ATP consumed: none.
- NAD^+ consumed: none.

4.3 Functional-group change

Glucose-6-phosphate is an aldose phosphate in its open-chain form. Fructose-6-phosphate is a ketose phosphate.

The net rearrangement moves the carbonyl functionality from C1 to C2 without changing the six-carbon skeleton or the position of the phosphate group.

$$\text{CHO}-(\text{CHOH})_4-\text{CH}_2\text{OPO}_3{}^{2-} \rightleftharpoons \text{CH}_2\text{OH}-\text{CO}-(\text{CHOH})_3-\text{CH}_2\text{OPO}_3{}^{2-}$$

This expression is a schematic connectivity comparison; the exact stereochemistry and hydroxyl positions are omitted.

4.4 Mechanistic sequence

1. The cyclic sugar opens to the linear form.
2. A catalytic base removes a proton adjacent to the carbonyl while the enzyme stabilizes the developing intermediate.
3. An enediol or enediolate-like intermediate is formed.
4. Proton transfer and carbonyl reformation produce the ketose.
5. The product can cyclize again.

4.5 Electron movement

The transformation is an intramolecular redox-neutral rearrangement. Electrons are redistributed within the sugar during the enediol-mediated isomerization.

There is no net transfer of electrons to NAD^+, and there is no net oxidation or reduction of the overall sugar molecule.

4.6 Why this step matters

Moving the carbonyl from C1 to C2 makes the later cleavage of fructose-1,6-bisphosphate possible in a way that produces two three-carbon phosphate intermediates.

Reference: Step 2. This is an isomerization, not a phosphorylation or oxidation reaction.

5 Step 3: Phosphorylation of Fructose-6-phosphate

5.1 Reaction

$$\boxed{\text{Fructose}-6\text{-phosphate} + \text{ATP} \xrightarrow[\text{Mg}^{2+}]{\text{Phosphofructokinase}-1} \text{Fructose}-1,6-\text{bisphosphate} + \text{ADP}}$$

Enzyme: Phosphofructokinase-1 (PFK-1).

5.2 Reactants and products

- Reactants: fructose-6-phosphate, ATP.
- Products: fructose-1,6-bisphosphate, ADP.
- ATP consumed: one molecule per glucose.
- Cofactor: Mg^{2+}.

5.3 Structural change

A phosphate group is added to the C1 hydroxyl group of fructose-6-phosphate.

The existing phosphate remains at C6.

$$F_6P + ATP \longrightarrow F_{1,6}BP + ADP$$

The product has two separate phosphate ester groups, one at C1 and one at C6. The prefix *bisphosphate* indicates that the two phosphate groups are attached at distinct positions.

5.4 Electron movement

The C1 hydroxyl oxygen attacks the terminal phosphorus of ATP. The enzyme facilitates transfer of the phosphoryl group, and ATP becomes ADP.

As in step 1, this is a phosphoryl-transfer reaction rather than an oxidation-reduction reaction.

5.5 Energy and regulation

This reaction consumes the second ATP molecule of glycolysis.

PFK-1 is a major regulatory enzyme. Its activity responds to the cell's energy state and other metabolic signals. The forward reaction is strongly favorable under typical cellular conditions.

Reference: Step 3. At the end of this step, two ATP molecules have been invested per original glucose.

6 Step 4: Cleavage of Fructose-1,6-bisphosphate

6.1 Reaction

$$\boxed{\text{Fructose}-1,6-\text{bisphosphate} \overset{\text{Aldolase}}{\rightleftharpoons} \text{Glyceraldehyde}-3\text{-phosphate} + \text{Dihydroxyacetone phosphate}}$$

Enzyme: Fructose-bisphosphate aldolase.

6.2 Reactants and products

- Reactant: fructose-1,6-bisphosphate, containing six carbons.
- Products: glyceraldehyde-3-phosphate (GAP) and dihydroxyacetone phosphate (DHAP), each containing three carbons.
- ATP consumed: none.
- NAD^+ consumed: none.

6.3 Carbon-skeleton cleavage

The six-carbon substrate is cleaved between C3 and C4.

$$\boxed{C_6 \longrightarrow C_3 + C_3}$$

The products are not chemically identical:

- GAP contains an aldehyde group in its open-chain form.
- DHAP contains a ketone group in its open-chain form.

Both products carry one phosphate group.

6.4 Mechanistic interpretation

The enzyme stabilizes the intermediates required for cleavage of the carbon-carbon bond. In class I aldolases, an active-site lysine forms a Schiff-base intermediate with the substrate. Class II aldolases instead use a metal ion to stabilize charged intermediates.

The detailed mechanism depends on the aldolase class.

6.5 Electron movement

This reaction redistributes bonding electrons as a carbon-carbon bond is cleaved. It is not a net oxidation-reduction reaction.

No carbon is lost as carbon dioxide. All six carbon atoms from glucose remain in the two three-carbon products.

Reference: Step 4. One molecule of glucose produces one GAP and one DHAP.

7 Step 5: Isomerization of DHAP to GAP

7.1 Reaction

$$\boxed{\text{Dihydroxyacetone phosphate} \overset{\text{Triose phosphate isomerase}}{\rightleftharpoons} \text{Glyceraldehyde}-3\text{-phosphate}}$$

Enzyme: Triose phosphate isomerase (TPI).

7.2 Reactants and products

- Reactant: DHAP.
- Product: GAP.
- ATP consumed: none.
- NAD^+ consumed: none.

7.3 Structural transformation

DHAP and GAP are three-carbon constitutional isomers. The carbonyl group changes from a ketone at C2 to an aldehyde at C1 in the open-chain representation.

$$HOCH_2{-}CO{-}CH_2OPO_3{}^{2-} \rightleftharpoons OHC{-}CHOH{-}CH_2OPO_3{}^{2-}$$

The formula above is schematic and suppresses protonation and stereochemical details.

7.4 Mechanism

1. An active-site base removes a proton from DHAP.
2. An enediol or enediolate intermediate is formed.
3. Proton transfer occurs.
4. Reformation of the carbonyl produces GAP.

The enzyme prevents undesirable side reactions and accelerates the reversible interconversion.

7.5 Electron movement

The electrons are redistributed within the three-carbon skeleton. There is no net electron transfer to NAD^+ and no net oxidation of the substrate.

7.6 Stoichiometric consequence

The GAP produced in step 4 is retained, and the DHAP is converted into a second GAP.

Therefore:

$$\boxed{1 \text{ glucose} \longrightarrow 2 \text{ GAP}}$$

Every subsequent reaction occurs twice per glucose molecule.

Reference: Step 5. The pathway now contains two equivalents of GAP.

8 Step 6: Oxidation and Phosphorylation of GAP

8.1 Reaction

$$\boxed{GAP + NAD^+ + P_i \overset{\text{GAPDH}}{\rightleftharpoons} 1{,}3{-}BPG + NADH + H^+}$$

Enzyme: Glyceraldehyde-3-phosphate dehydrogenase (GAPDH).

8.2 Reactants and products

For each GAP molecule:

- Reactants: GAP, NAD^+, inorganic phosphate.
- Products: 1,3-bisphosphoglycerate, NADH, H^+.
- ATP consumed: none.
- NAD^+ reduced: one molecule.

Because two GAP molecules are processed per glucose, the net contribution of this step is:

$$2\,\mathrm{GAP} + 2\,\mathrm{NAD^+} + 2\,\mathrm{P}_i \longrightarrow 2\,\mathrm{1{,}3{-}BPG} + 2\,\mathrm{NADH} + 2\,\mathrm{H^+}$$

8.3 Structural transformation

The aldehyde carbon of GAP is oxidized to the acyl-phosphate carbon of 1,3-BPG.

$$\mathrm{OHC{-}CHOH{-}CH_2OPO_3{}^{2-}}$$

$$\Downarrow$$

$$\mathrm{O{=}C(OPO_3{}^{2-}){-}CHOH{-}CH_2OPO_3{}^{2-}}$$

These are schematic open-chain representations. The first phosphate in the product is an acyl phosphate; the second is the pre-existing phosphate ester.

8.4 Where the phosphate comes from

The phosphate attached at C3 of GAP is already present before this reaction. The new phosphate at C1 comes from inorganic phosphate, P_i.

It does **not** come from ATP.

8.5 Electron transfer

This is the principal oxidation-reduction step of glycolysis.

1. The aldehyde group of GAP is oxidized.
2. Electrons are transferred to NAD^+ through the enzyme's catalytic mechanism.
3. NAD^+ accepts a hydride equivalent, corresponding to two electrons and one proton, to form NADH.
4. An additional proton may be released to solution, depending on the chosen reaction convention.

The simplified redox bookkeeping is:

$$\boxed{\mathrm{NAD^+} + 2\,\mathrm{H} \longrightarrow \mathrm{NADH} + \mathrm{H^+}}$$

Here, the notation represents net reducing-equivalent transfer, not the presence of free atomic hydrogen as a reaction intermediate.

The GAPDH mechanism uses a catalytic cysteine residue to form a covalent intermediate with the aldehyde-derived substrate. Oxidation generates a high-energy acyl intermediate, which is then attacked by inorganic phosphate to form the acyl phosphate of 1,3-BPG.

8.6 Energy capture

The energy released during oxidation is conserved in two forms:

- The reduced electron carrier NADH.
- The high phosphoryl-transfer potential of 1,3-BPG.

This is an example of coupling oxidation to formation of a high-energy phosphate compound.

Reference: Step 6. Per glucose, two NADH molecules are produced. No ATP is produced directly in this reaction.

9 Step 7: ATP Production from 1,3-BPG

9.1 Reaction

$$\boxed{\text{1,3}-\text{BPG} + \text{ADP} \overset{\text{Phosphoglycerate kinase}}{\rightleftharpoons} \text{3-PG} + \text{ATP}}$$

Enzyme: Phosphoglycerate kinase (PGK).

9.2 Reactants and products

For each 1,3-BPG molecule:

- Reactants: 1,3-BPG, ADP.
- Products: 3-phosphoglycerate, ATP.
- ATP produced: one molecule.

Per glucose:

$$2\,\text{1,3}-\text{BPG} + 2\,\text{ADP} \longrightarrow 2\,\text{3-PG} + 2\,\text{ATP}$$

9.3 Phosphate transfer

The phosphate attached to C1 of 1,3-BPG is transferred to ADP.

$$\boxed{\text{C1 acyl phosphate} \longrightarrow \text{ADP} \longrightarrow \text{ATP}}$$

The phosphate at C3 remains attached to the three-carbon skeleton. The product is 3-phosphoglycerate.

9.4 Electron movement

This is a phosphoryl-transfer reaction, not an oxidation-reduction reaction.

The reaction does not transfer electrons to NAD^+, and the carbon oxidation state does not undergo a net redox change.

The enzyme facilitates phosphoryl transfer from the high-energy acyl phosphate to ADP.

9.5 Energy accounting

This is the first ATP-producing reaction of glycolysis.

Because two molecules of 1,3-BPG are generated per glucose, this step produces two ATP molecules per glucose.

The process is called **substrate-level phosphorylation**: ATP is formed by direct transfer of a phosphoryl group from a metabolic intermediate to ADP.

Reference: Step 7. Two ATP molecules are generated per glucose. This compensates for the two ATP molecules consumed in steps 1 and 3, but does not yet create a net gain.

10 Step 8: Phosphate Migration

10.1 Reaction

$$\boxed{\text{3-PG} \overset{\text{Phosphoglycerate mutase}}{\rightleftharpoons} \text{2-PG}}$$

Enzyme: Phosphoglycerate mutase.

10.2 Reactants and products

- Reactant: 3-phosphoglycerate.
- Product: 2-phosphoglycerate.
- ATP consumed or produced: none.
- NAD^+ consumed: none.

10.3 Structural transformation

The phosphate group changes position on the three-carbon skeleton.

$$^{-}OOC{-}CH(OH){-}CH_2OPO_3{}^{2-}$$

$$\Downarrow$$

$$^{-}OOC{-}CH(OPO_3{}^{2-}){-}CH_2OH$$

The phosphate moves from C3 to C2.

10.4 Mechanistic sequence

In the common phosphohistidine-dependent mechanism:

1. The enzyme transfers a phosphate group to C2, forming a transient bisphosphorylated intermediate.
2. The original C3 phosphate is removed.
3. The result is 2-phosphoglycerate, with phosphate attached at C2.

Some organisms use mechanistically distinct phosphoglycerate mutases, but the net transformation is the same.

10.5 Electron movement

This is an intramolecular rearrangement mediated by the enzyme. No net electron transfer to NAD^+ occurs.

10.6 Why the phosphate moves

Moving the phosphate to C2 prepares the molecule for dehydration in step 9. The resulting phosphoenolpyruvate has a high phosphoryl- transfer potential.

Reference: Step 8. The phosphate is relocated; ATP is neither consumed nor generated.

11 Step 9: Dehydration to Phosphoenolpyruvate

11.1 Reaction

$$\boxed{\text{2-PG} \overset{\text{Enolase}}{\rightleftharpoons} \text{PEP} + \text{H}_2\text{O}}$$

Enzyme: Enolase.

11.2 Reactants and products

- Reactant: 2-phosphoglycerate.
- Products: phosphoenolpyruvate (PEP), water.
- ATP consumed or produced: none.
- NAD^+ consumed: none.

11.3 Structural transformation

The reaction removes the elements of water from 2-phosphoglycerate and creates a carbon-carbon double bond.

$$^{-}\text{OOC}-\text{CH}(\text{OPO}_3{}^{2-})-\text{CH}_2\text{OH}$$

$$\Downarrow \quad -\text{H}_2\text{O}$$

$$^{-}\text{OOC}-\text{C}(\text{OPO}_3{}^{2-})=\text{CH}_2$$

The displayed formulas are schematic representations of the connectivity of the substrate and product.

11.4 Bond changes

1. A proton is removed from one carbon.
2. The hydroxyl group is eliminated from the adjacent carbon.
3. The electrons are redistributed to form the C=C double bond.
4. Water is released.
5. The product is the enol phosphate PEP.

11.5 Electron movement

The enzyme promotes an elimination reaction. Bond electrons are redistributed within the carbon skeleton, resulting in the double bond of PEP.

There is no net transfer of electrons to NAD^+ and no direct ATP formation in this step.

11.6 Energy significance

PEP has a high phosphoryl-transfer potential. Its conversion to pyruvate is strongly favorable because the initial enol product can tautomerize to the more stable keto form of pyruvate.

Reference: Step 9. Dehydration creates the high-energy enol phosphate used in step 10.

12 Step 10: ATP Production and Formation of Pyruvate

12.1 Reaction

$$\boxed{\mathrm{PEP} + \mathrm{ADP} + \mathrm{H^+} \xrightarrow[\mathrm{Mg^{2+}, K^+}]{\text{Pyruvate kinase}} \text{Pyruvate} + \mathrm{ATP}}$$

Enzyme: Pyruvate kinase.

12.2 Reactants and products

For each PEP molecule:

- Reactants: PEP, ADP.
- Products: pyruvate, ATP.
- ATP produced: one molecule.
- Required ions commonly include Mg^{2+} and K^+.

Per glucose:

$$2\,\mathrm{PEP} + 2\,\mathrm{ADP} + 2\,\mathrm{H^+} \longrightarrow 2\,\mathrm{Pyruvate} + 2\,\mathrm{ATP}$$

The exact proton coefficient depends on the protonation conventions used for the metabolites.

12.3 Phosphate transfer

The phosphate group of PEP is transferred to ADP.

$$\boxed{\mathrm{PEP} \longrightarrow \mathrm{ADP} \longrightarrow \mathrm{ATP}}$$

The three-carbon skeleton becomes pyruvate.

12.4 Electron movement and tautomerization

The initial product of phosphoryl transfer is an enol form of pyruvate. It undergoes keto-enol tautomerization to produce the more stable keto form.

The electron redistribution associated with this tautomerization helps make the overall reaction strongly favorable.

This is not an additional oxidation of pyruvate. Rather, the reaction couples phosphoryl transfer with formation of the thermodynamically favored keto product.

12.5 Energy accounting

This is the second ATP-producing reaction of glycolysis.

Two PEP molecules are produced per glucose, so this step generates two ATP molecules per glucose.

Reference: Step 10. The final glycolytic product is two pyruvate molecules, and this step produces the final two ATP molecules of the pathway.

13 Complete Reaction Table

Step	Enzyme	Main transformation	ATP	Redox
1	Hexokinase	Glucose → glucose-6-phosphate	−1	No
2	Phosphoglucose isomerase	G6P → F6P	0	No
3	Phosphofructokinase-1	F6P → fructose-1,6-bisphosphate	−1	No
4	Aldolase	FBP → GAP + DHAP	0	No
5	Triose phosphate isomerase	DHAP → GAP	0	No
6	GAP dehydrogenase	GAP → 1,3-BPG; NAD^+ reduced	0	Yes
7	Phosphoglycerate kinase	1,3-BPG → 3-PG	+2 per glucose	No
8	Phosphoglycerate mutase	3-PG → 2-PG	0	No
9	Enolase	2-PG → PEP + water	0	No
10	Pyruvate kinase	PEP → pyruvate	+2 per glucose	No

ATP accounting:

$$\begin{aligned} \text{ATP consumed} &= 2, \\ \text{ATP produced} &= 4, \\ \text{Net ATP} &= 4 - 2 = 2. \end{aligned}$$

NADH accounting:

$$\boxed{1 \text{ glucose} \longrightarrow 2 \text{ NADH}}$$

14 Phosphate-Transfer Map

The pathway uses two distinct sources of phosphate.

14.1 Phosphate groups donated by ATP

1. Step 1: ATP donates a phosphate to glucose at C6.
2. Step 3: ATP donates a phosphate to fructose-6-phosphate at C1.

These are the two ATP-investment reactions.

14.2 Phosphate groups donated by inorganic phosphate

1. Step 6: inorganic phosphate is incorporated into 1,3-BPG during the oxidation of GAP.

The phosphate incorporated at C1 of 1,3-BPG is not donated by ATP.

14.3 Phosphate groups transferred to ADP

1. Step 7: 1,3-BPG transfers its C1 phosphate to ADP.
2. Step 10: PEP transfers its phosphate to ADP.

Both reactions produce ATP by substrate-level phosphorylation.

14.4 Simplified phosphate-flow diagram

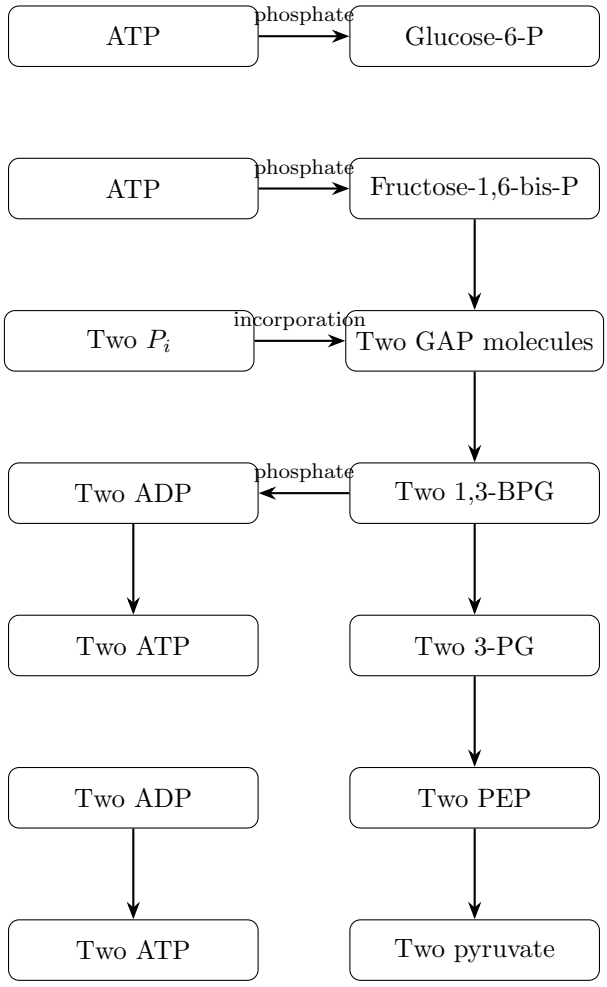


Figure 2: Schematic accounting for phosphate investment and transfer. The map is an accounting aid rather than a literal spatial model of enzyme-bound intermediates.

15 Electron-Transfer Map

Only step 6 directly reduces NAD^+ during glycolysis.

$$\boxed{2\,\mathrm{GAP} + 2\,\mathrm{NAD^+} + 2\,\mathrm{P}_i \longrightarrow 2\,\mathrm{1{,}3{-}BPG} + 2\,\mathrm{NADH} + 2\,\mathrm{H^+}}$$

The important distinction is between oxidation of the carbon substrate and phosphoryl transfer:

- Step 6 oxidizes GAP and reduces NAD^+.
- Step 7 transfers a phosphate group to ADP.
- Step 10 transfers another phosphate group to ADP.

No molecular oxygen is consumed directly in the ten reactions of glycolysis. Nevertheless, glycolysis contains an oxidation reaction because NAD^+ acts as an electron acceptor.

15.1 What happens to the NADH?

NADH must eventually be reoxidized to NAD^+ if glycolysis is to continue.

Under aerobic conditions, the reducing equivalents can be transferred to mitochondrial oxidative metabolism through appropriate shuttle systems.

Under anaerobic conditions in animal cells, lactate dehydrogenase can reduce pyruvate to lactate:

$$\text{Pyruvate} + \text{NADH} + \text{H}^+ \rightleftharpoons \text{Lactate} + \text{NAD}^+.$$

This reaction regenerates NAD^+, allowing GAPDH to continue operating. It is a subsequent reaction, not one of the ten reactions of glycolysis itself.

16 Overall Stoichiometry

The overall glycolytic reaction is conventionally written as

$$\boxed{\begin{aligned}&\text{Glucose} + 2\,\text{NAD}^+ + 2\,\text{ADP} + 2\,\text{P}_i \\ &\qquad\longrightarrow 2\,\text{Pyruvate} + 2\,\text{NADH} + 2\,\text{H}^+ + 2\,\text{ATP} + 2\,\text{H}_2\text{O}.\end{aligned}}$$

This equation uses a common biochemical protonation convention. The exact coefficient for protons depends on how ionic species are represented.

16.1 Net yield per glucose

Quantity	**Net yield**
Glucose consumed	1
Pyruvate produced	2
ATP consumed	2
ATP produced	4
ATP net gain	2
NADH produced	2
Carbon dioxide produced directly	0

16.2 Carbon balance

$$\boxed{\text{C}_6\text{H}_{12}\text{O}_6 \longrightarrow 2\,\text{C}_3\text{-carbon products}}$$

The notation above emphasizes carbon accounting rather than serving as a complete balanced chemical equation.

The six glucose carbon atoms remain in the two pyruvate molecules. No carbon is lost as carbon dioxide during glycolysis.

17 Key Biochemical Conclusions

1. Glycolysis consists of ten enzyme-catalyzed reactions.
2. The first five reactions consume two ATP molecules and generate two GAP molecules.
3. The last five reactions occur twice per original glucose.
4. GAP is oxidized in step 6, generating two NADH molecules per glucose.
5. Phosphate transfer to ADP occurs in steps 7 and 10, generating four ATP molecules in total.
6. The net ATP gain is two ATP molecules per glucose.
7. Steps 1, 3, and 10 are the principal strongly favorable regulatory reactions under typical cellular conditions.
8. Glycolysis does not directly consume molecular oxygen, and it does not produce carbon dioxide.
9. NAD^+ must be regenerated for glycolysis to continue.
10. FAD is not a reactant in any of the ten canonical glycolytic reactions.

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